

Euclidean Geometry In Mathematical Olympiads

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Problem 1.38



geometry

cyclic quadrilateral

incenter

Chapter 1

fgm

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#1

Problem 1.38

In the cyclic quadrilateral $ABCD$, let I_1 and I_2 denote the incenters of $\triangle ABC$ and DBC , respectively. Prove that I_1I_2BC is cyclic.

This post has been edited 1 time. Last edited by 277546, Mar 8, 2018, 10:24 PM

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#2

Solution



By The Incenter/Excenter Lemma, I_2BC has a circumcircle with center L , where L is the midpoint of arc BC . Similarly, I_1BC does as well. Thus, I_1I_2BC is cyclic.

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#3

Wait, do we have to address the case in which B is opposite A as well? Because the hints seem to refer to that.

ythomashu

6256 posts

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#4

I don't see how that changes anything

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#5

Well L wouldn't be the midpoint of the same arc, it would be of a different one (i think)

WizardMath

2487 posts

Mar 9, 2018, 12:50 AM • 8

#7

The fact that $ABCD$ is given in this order implies that the only configuration you need to take care of is the one you used to prove. The other configuration doesn't have the problem statement true.

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#8

Huh. Thats what I figured, so I was quite confused with the hints. Maybe it's a typo?

HardyRamanuj...

71 posts

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#9

So i just solved this problem but I have solved the problem for the case in which A and B are opposite (proved $\angle I_2CI_1 = \angle I_2BI_1$ (simple angle chase))and proved that the statement is wrong if they are adjacent using the fact that $\angle BIC = 90 + \angle A/2$. But after seeing this , I am really confused.

Arthur.

133 posts

Jun 17, 2018, 5:31 PM • 2

#10



HardyRamanujan wrote: